

Book Chapter

Chapter Title:

**Explicit vs. Implicit Coordination Mechanisms and Task Dependencies:
One Size Does Not Fit All**

J. Alberto Espinosa
American University

F. Javier Lerch
Robert E. Kraut
Carnegie Mellon University

Book Title:

**Team Cognition:
Process and Performance at the Inter and Intra-Individual Level**

Editors: S. M. Fiore, E. Salas and J. A. Cannon-Bowers

Espinosa, J. A., Lerch, F. J. and Kraut, R. E. (2004). Explicit vs. Implicit Coordination Mechanisms and Task Dependencies: One Size Does Not Fit All. Team Cognition: Understanding the Factors that Drive Process and Performance. E. Salas and S. M. Fiore. Washington, D.C., American Psychological Association: 107-129, ch. 6, ISBN 1-59147-103-6.

This research was partially supported by the National Science Foundation, Grant ISS-9812123
and by Lucent Technologies

Team Cognition, Task Dependencies and Coordination: One Size Does Not Fit All

Introduction

Teams are important units of organizational work because they bring diverse expertise, skills and resources to complex tasks that may be too large or complex for a single individual to undertake. However, as projects and teams grow in size and complexity, tasks and member dependencies become more numerous and more complex, thus increasing the need for team coordination. Effective teams manage these dependencies using a number of explicit and implicit coordination mechanisms and processes. Teams coordinate explicitly using task programming mechanisms (e.g., schedules, plans, procedures, etc.) or by communicating (e.g., orally, in writing, formally, informally, interpersonally, in groups). We call these mechanisms “explicit” because team members use them purposely to coordinate. However, teams can also coordinate “implicitly” (i.e., without consciously trying to coordinate) through team cognition, or knowledge that team members share about the task and about each other. This shared knowledge helps team members understand what is going on with the task and anticipate what may happen next and which actions team members are likely to take, thus helping them coordinate. In this chapter we discuss the interplay between explicit and implicit coordination mechanisms and how they jointly affect team coordination and performance.

Because dependencies can be managed in more than one way, teams will make decisions on which explicit coordination mechanisms to use. However, the use of such explicit mechanisms both influence (e.g., by interacting, by sharing documents, etc.) and are influenced by the existing level of team cognition (e.g., members with shared knowledge may communicate less frequently and more efficiently). Consequently, it is important that we understand how explicit and implicit

coordination mechanisms complement and interact with each other. We also discuss in this chapter the need to understand the nature of the multiple dependencies involved in a task in order to figure out which mechanisms can be more effective in helping teams coordinate. Different mechanisms, whether explicit or implicit, will have varying degrees of effectiveness for different tasks and for different stages of a given task. For example, team communication may be very important for complex intellectual tasks in which task dependencies are somewhat uncertain or for the early stages of a task, when team members are still unfamiliar with the task and with each other. On the other hand, team communication may not be so important for more mechanical tasks (e.g., assembly line) in which dependencies are more predictable or for the late stages of a task, once team members know each other or once they have implemented division of labor schemes.

Throughout this chapter we focus on asynchronous (i.e., non real-time) and geographically dispersed task contexts. This is an important distinction because most of the literature on team cognition has focused on synchronous (i.e., same-time) and co-located teams (e.g., flight crews, medical emergency units, etc.). However, work arrangements and the nature of the resulting dependencies will vary substantially depending on whether the team is separated by time or by distance. We begin our discussion by describing a unified framework of team coordination that incorporates both explicit and implicit coordination mechanisms. Next we discuss the importance of identifying key dependencies and coordination types for a given task. Finally, we summarize our conclusions. Throughout our discussion, we present examples and results from our empirical studies with: (1) decision [asynchronous] teams who managed simulated companies for Carnegie Mellon University's Management Game course (Espinosa, Carley et al. 2002); and (2) large-scale, geographically distributed software development teams from a Fortune 500 telecommunications company (Espinosa, Kraut et al. 2001; Espinosa 2002; Espinosa, Kraut et al. 2002). We refer in

this chapter to the first teams as the “decision teams” and to the second teams as the “software teams”.

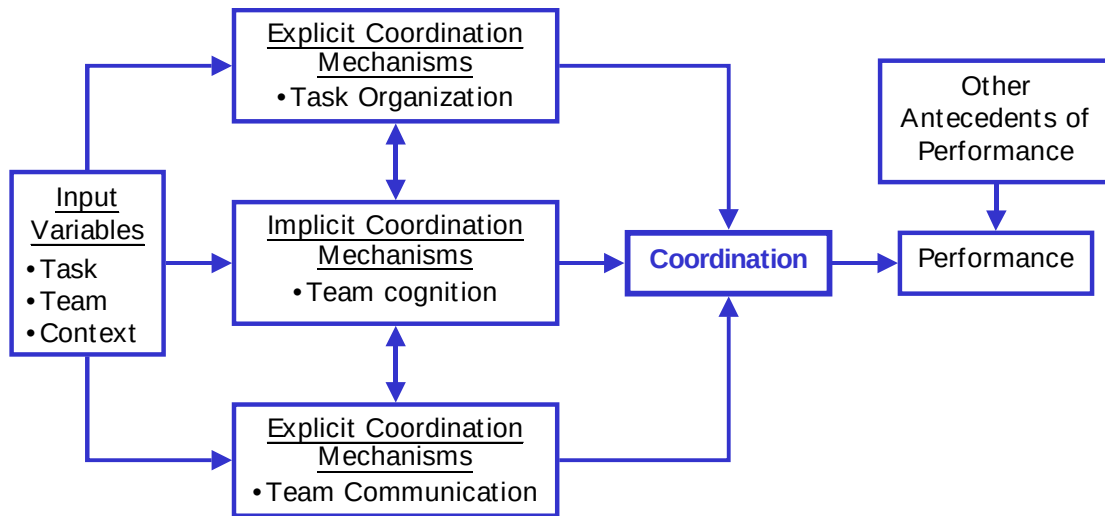


Figure 1: Theoretical Framework

Theoretical Framework

In this section of the chapter we propose and describe a theoretical framework (illustrated in Figure 1) to study the effects of team cognition on team coordination and performance. This framework has three important properties. First, it is an input-process-output (IPO) model, which is appropriate for the study of teams (Kraemer and Pinsonneault 1990; McGrath 1991; McGrath and Hollingshead 1994). Input variables in the framework represent task, team and context factors that give rise to different work arrangements and dependencies. Teams manage these dependencies using a mix of coordination mechanisms. The effective management of these dependencies influences the team’s state of coordination, which in turn affects team performance. Second, the framework includes both, explicit and implicit coordination mechanisms which, as we discuss later in the chapter, need to be modeled jointly because they may complement, affect or interact with

each other in their effect on team coordination. Finally, the framework relates to how task dependencies originate (i.e., the need to coordinate) and how these dependencies are managed (i.e., the act of coordinating). In this section of the chapter, we provide the theoretical foundations of the framework. We first define coordination and then describe the framework by discussing: how coordination affects performance; how implicit and explicit coordination mechanisms affect team coordination; and how input variables (i.e., task, team and context) influence the mix of coordination mechanisms employed by a team. We then offer an integrated view of our theoretical framework. We discuss the third property of the framework in the next section of this chapter in which we relate the different elements of the framework to the underlying task dependencies.

Coordination Defined: Managing Dependencies

We draw from the research literature on coordination theory to define coordination as the effective management of dependencies among sub-tasks, resources (e.g., equipment, tools, etc.) and people (Malone and Crowston 1990; 1994). If things can be done totally independently, then there is nothing to coordinate. Conversely, when multiple individuals, sub-tasks and resources need to interact in a synchronized fashion to carry out a joint task, it gives rise to dependencies among them. Team members can perform their individual responsibilities competently and still be very uncoordinated with the rest of the team if the respective dependencies among their sub-tasks are not well managed. We use this view of coordination from the perspective of managing dependencies throughout this chapter to explain how different elements of the proposed framework fit together. This definition of coordination is very useful when conducting research because it helps define concepts and develop measures. For example, coordination can be viewed as both, a process (i.e., coordinating) and an outcome (i.e., coordination success). The process of coordinating can be defined as “the activities carried out by team members when managing dependencies”. For

instance, when a software team convenes a debriefing meeting so that all parties can share the status of their respective tasks, they are managing the dependencies involved in developing software as a team by communicating with each other about what is going on with their individual task assignments. Coordination as an outcome (i.e., coordination success) can be defined as “the extent to which dependencies have been effectively managed”. For example, a software team will be well coordinated if all software parts for a given project work well together, are delivered on schedule, and are produced according to the established software process (i.e., technical, temporal and software process dependencies have been effectively managed).

Coordination: An Antecedent of Team Performance

Performance is an outcome of utmost interest in team research. Team coordination is useful to the extent that it leads to team performance, but coordination is not always important for performance. For example, coordination may not be so important for team problem-solving tasks in which a “eureka” type solution exists, which is evident to all members when one member figures it out. If members of a team can work independently and if there are no task dependencies, then there is nothing to coordinate and, consequently, coordination will not affect performance (Thompson 1967; VanDeVen, Delbecq et al. 1976; Malone and Crowston 1994). In contrast, more complex tasks with substantial dependencies can indeed benefit from coordination. However, even if coordination is necessary for performance, it may not always be sufficient to ensure performance. First of all, some task dependencies are more critical to performance than others. A team may be doing a great job at managing some task dependencies that are important but not critical for task performance (e.g., delivering software on time) while not paying much attention to other dependencies that are critical for task performance (e.g., delivering high quality and error-free software). For instance, in our study with software teams some technical and project managers

described situations in which they had implemented software product features in a very effective and productive manner, only to find out later on that these features had been removed from the product because they were not what the client wanted or what the market needed. Similar “feature churn” problems have been documented in other studies (Crowston and Kammerer 1998). Similarly, in our study with decision teams we found that the effect of task coordination (e.g., avoiding duplication of work, having clear assignment of tasks, etc.) on performance is indirect, mediated by functional strategy coordination (e.g., finance strategy well coordinated with production strategy).

Second, there may be other antecedents of performance that are unrelated to coordination, which may hinder or enhance performance. For example, in our study with decision teams we found some evidence that strategy coordination had a positive effect on team performance, but it only explained a small percentage of variance. Instructors of the Management Game simulation course indicated that in addition to having coordinated strategies, successful teams also have sound functional strategies (i.e., finance, marketing and operations) and a good understanding of the competitive environments in which their simulated firms operate.

Explicit Coordination Mechanisms: The Classic Organizational Theory View

A coordination mechanism is one that helps teams manage dependencies. For example, simple things in our daily lives like a traffic light and a flight schedule can be viewed as coordination mechanisms because they help us manage our dependencies with other drivers and the airlines, respectively. Consequently, explicit coordination mechanisms can be defined as those mechanisms explicitly employed by a team for the purpose of managing task dependencies. Explicit coordination mechanisms and processes have been studied in the classical organizational research literature for several years. This literature suggests that teams coordinate explicitly by

using task organization mechanisms or by communicating. March and Simon suggested that teams use task organization (or “task programming”) mechanisms for the most routine aspects of the task because the respective dependencies are more predictable, and thus can be more easily managed in a programmed way (March and Simon 1958; Thompson 1967). Examples of these mechanisms include things like division of labor, tools, schedules, plans, manuals and specifications.

When routines change or when the task has very little or no routine aspects, task organization mechanisms are less effective because dependencies can no longer be managed in a programmed way. March and Simon suggested that teams resort to communication (i.e., “coordination by feedback”) in such cases. For example, a software project schedule may be rendered useless if several deliverable deadlines have already been missed or if there has been a crisis (e.g., a major hardware failure) that calls for re-scheduling. When these things happen the team needs to communicate to cope with the changing situation or to implement different coordination mechanisms more suitable to the new situation. Coordination by communication can be interpersonal or in groups. It can also be formal or informal. For instance, an empirical study with software teams at an organization found that teams not only coordinated by communicating formally via meetings and documents, but that they also did a substantial amount of coordination informally during encounters in public places like water coolers and hallways (Kraut and Streeter 1995).

This discussion underscores the fact that different coordination mechanisms may be more suitable for different tasks. Furthermore, the same task may require the use of different coordination mechanisms over time. Studies conducted at Carnegie Mellon have analyzed software student teams developing small applications for external clients and found that these teams changed the mix of coordination mechanisms they used over time (Wholey, Kiesler et al. 1996). For

example, Wholey et al. found that teams communicate more intensely at the beginning. Interestingly, the study also found: that successful teams only communicated moderately toward the end; that unsuccessful novice teams communicated too little, while unsuccessful expert teams communicated too much; and that division of labor was an effective task organization mechanism only toward the end of the task, once team members knew each other's skills well.

Implicit Coordination Mechanisms: A More Recent View from Team Cognition Research

More recent research in team cognition suggests that as team members interact with each other and gain expertise with the task, they develop shared knowledge about the task and the team, which helps them coordinate implicitly (Cannon-Bowers, Salas et al. 1993; Klimoski and Mohammed 1994; Levesque, Wilson et al. 2001). For example, team members coordinate implicitly by providing necessary information to each other without having to ask (Kleinman and Serfaty 1989; Entin and Serfaty 1999). Such implicit coordination has been referred to as the "synchronization of member actions based on unspoken assumptions about what others in the group are likely to do" (Wittenbaum and Stasser 1996). Consequently, we define implicit coordination mechanisms as those that are available to members from team cognition, which enable them to explain and anticipate task states and member actions, thus helping them manage task dependencies.

There are a number of different streams of research that have studied these implicit coordination mechanisms. A recent survey (Cannon-Bowers and Salas 2001) identified over 20 team cognition labels and constructs described and studied in the literature, including shared (or team) mental models (Rouse and Morris 1986; Rouse, Cannon-Bowers et al. 1992; Cannon-Bowers, Salas et al. 1993; Klimoski and Mohammed 1994; Kraiger and Wenzel 1997), team situation awareness (Wellens 1993; Endsley 1995), transactive memory (Wegner 1986; Liang,

Moreland et al. 1995), mutual knowledge (Krauss and Fussell 1990; Fussell and Krauss 1992; Cramton 2001) and collective mind (Weick and Roberts 1993). These constructs are conceptually distinct, but they share the commonality that some form of knowledge similarity or knowledge complement helps team members: explain other members' actions, understand what is going on with the task; develop accurate expectations about future member actions and task states; and communicate meaning efficiently because of shared vocabulary and more common ground. All of this improves coordination.

Team cognition research has focused primarily on developing theoretical foundations, but there has been a paucity of empirical studies. Fortunately, this is beginning to change. For example, a recent study of dyads working in a flight simulation task found that shared mental models have a positive, but indirect effects on team performance, mediated by team process (e.g., coordination) (Mathieu, Goodwin et al. 2000). Also, most of the research in shared cognition has focused on real-time and co-located teams. But because time and distance separation provides fewer opportunities for members to interact and acquire shared knowledge, team cognition is much more difficult to develop in asynchronous and geographically dispersed environments, thus the importance of conducting research in these contexts. Fortunately, new research is beginning to emerge in these contexts. For instance, a recent study of large-scale software developers found that “administrative” (i.e., explicit) coordination positively affects team performance, but it also found that knowing where expertise is located in the team has a positive and significant effect on team effectiveness and efficiency, above and beyond the effects of administrative coordination (Faraj and Sproull 2000). Another study conducted with consulting teams composed of MBA students working on three-month long projects in actual client organizations found preliminary evidence that transactive memory is positively correlated with team coordination (Lewis 2000). Yet another

study with software requirements teams at two organizations found that explicit management of dependencies helped understand why teams were coordinated, but only up to a certain point, and that teams that exhibited a “collective mind”—i.e., “individuals developing a shared understanding of the group’s task and each other” (Weick and Roberts 1993)—were more coordinated (Crowston and Kammerer 1998).

Our research with decision teams has also found that shared mental models have a positive effect on task coordination (i.e., management of general task dependencies—e.g., avoiding duplication of work, tasks are clearly assigned, etc.) and strategy coordination (i.e., management of functional strategy dependencies—e.g., finance, operations and marketing strategies) (Espinosa, Carley et al. 2002). Observations in a related study with two distributed decision teams (i.e., students from a Mexican university and from Carnegie Mellon University working on the same team) suggested that these teams were uncoordinated at the beginning, but that they became progressively more coordinated over time as they became more familiar with the task and with each other¹. Analysis from our field research with software teams has also found evidence that organized shared knowledge (i.e., shared mental models) of key concepts, processes and products, knowing who knows what in the team (i.e., transactive memory), and knowing what is happening with the task (i.e., task awareness) and who is around (i.e., presence awareness) help teams coordinate, particularly when teams are geographically distributed (Espinosa, Kraut et al. 2001; Espinosa 2002).

In sum, as team members develop experience with the task and interact with each other they develop team cognition that helps them coordinate implicitly. But team communication and the use of task organization mechanisms affect how team cognition develops. In turn, the strength and type

¹ Students from each site visited the other site twice during the semester and also had video conference meetings.

of team cognition developed influences which task organization mechanisms are used by the team (and how they are used) and how they communicate.

Input Variables: Effect on Mix of Implicit and Explicit Coordination Mechanisms

As the following discussion suggests, a team will employ a mix of coordination mechanisms that it deems most suitable to manage task dependencies in a given context. As Table 1 shows, task, team and context variables influence the number, complexity and types of dependencies that are present in the task and, consequently, which coordination mechanisms are used by team members to manage these dependencies. For example, decision and software teams in our studies faced different types of dependencies. Decision teams had dependencies on general task activities (e.g., avoiding duplicate work, task assignments) and functional strategy dependencies (e.g., synchronizing finance and operations strategies), whereas software teams faced technical (i.e., software parts working well together), temporal (i.e., parallel activities completed on time) and software process dependencies (i.e., activities carried out according to the formal process).

Task	• Decision making	• Large-scale software development for telecommunications products
Teams	• Simulated management teams	• Software teams
Dependencies	• General task activities—clear assignment of tasks, no redundancies • Among functional strategies—integration of finance, marketing and operational strategies	• Technical—integration of software parts • Temporal—per schedule • Software Process—according to the established software process
Coordination	• Activity and functional strategy coordination	• Technical, temporal and process coordination
Complexity	• Moderate	• Very high
Context	• Same place • Different time	• Different place • Different time

Table 1: Tasks, Teams and Dependencies

Team variables also influence the types of dependencies present. For instance, larger teams are likely to have more numerous and complex dependencies than smaller teams. Also, teams with

a long history of working together may be more experienced at managing team member dependencies. Finally, context variables also influence the types of dependencies present in the task. Things like technology and organizational processes can impose structure and restrictions that affect how teams interact and carry out their tasks. For example, teams that are co-located are likely to use a different mix of coordination mechanisms than teams who are geographically distributed and whose interaction is mediated by communication technologies. Similarly, teams working in real-time contexts (e.g., flying an airplane, operating on a patient) are likely to use a different mix of coordination mechanisms than teams working on asynchronous tasks (e.g., strategic planning, large-scale software development). We will now explore in more detail how input variables change the nature of dependencies.

An Integrated Framework: Explicit and Implicit Coordination

Our discussion suggests the integration of the classical organizational view of team coordination with newer views from team cognition research. As we discuss next, every task is different, and the nature of dependencies present influences which mix of coordination mechanisms can better help teams coordinate. Teams working on familiar and routine tasks and larger teams may benefit more from task organization mechanisms (i.e., plans, schedules, programs, tools, etc.). Teams working on less familiar tasks and in uncertain conditions and smaller teams may benefit more by coordinating via communication. In addition, as the task progresses and as team members interact over time, their team cognition will gradually become stronger and, while team members may continue to use some explicit coordination mechanisms, they will probably begin to substitute other explicit mechanisms with implicit mechanisms. Explicit mechanisms that consume more time and effort (e.g., formal meetings, written reports) are more likely to be substituted, particularly when teams are under time pressure.

Consequently, team studies that focus on explicit coordination mechanisms need to take into account team members' use of implicit coordination mechanisms. Furthermore, the interplay between explicit and implicit coordination mechanisms may give rise to complementary and interaction effects. For example, the effect that a new software version management system may have on coordination in software development may be different for a new team than for one who has been working together in the same task domain for several years. For example, the introduction of a new software modeling tool had noticeably different effects on teams at two sites we studied. Team members at one site had a long history of working together and little turnover, and had developed paradigms and methods deeply rooted in tradition. This group had great difficulty adapting to the new tool, and many indicated that the tool did not help them coordinate. The team at the other site had only been in existence for approximately three years and had a high turnover rate. Most developers at this other site indicated that the new tool helped them coordinate. Similarly, studies of team coordination that focus on team cognition need to take into account the team's use of explicit coordination mechanisms. For example, a software team that does not use a configuration management system for software version control may need to rely more on shared task knowledge than a team that uses such systems. In sum, we suggest the use of an integrated framework of team coordination that incorporates both implicit and explicit coordination mechanisms, as illustrated in Figure 1.

Different Types of Dependencies and Coordination: One Size Does Not Fit All

In a nutshell, the framework proposed above suggests that task, team and context variables influence the nature of dependencies present in a task, which in turn affect the mix of coordination mechanisms necessary to manage these dependencies effectively. To the extent that these dependencies are important for performance, coordination will affect team performance. However,

as suggested in other studies, figuring out which coordination mechanisms are most effective in managing the various dependencies requires an understanding of these dependencies (Crowston and Kammerer 1998). While some argue that shared mental models or transactive memory can improve coordination and performance, we argue that there may be important complementarities, tradeoffs and interactions between implicit and explicit mechanisms. Consequently, if key dependencies can be managed effectively with explicit coordination mechanisms alone, then team cognition may not improve coordination noticeably. For example, configuration management systems (i.e., version control systems) in large-scale software projects use a library metaphor in which software is “checked out”, updated and then “checked in”, thus facilitating parallel development by multiple programmers, who may not even know each other. Modern configuration management systems automatically protect all parts of the software code that are affected by the software part that was checked out so that developers don’t interfere with each other, and also contain features to manage workflow (Grinter, Herbsleb et al. 1999). Such systems make communication and team cognition less important when coordinating small modifications to the software (Espinosa 2002). However, these tools may not be sufficient to coordinate larger modifications because dependencies are more complex, so communication and team cognition can indeed provide incremental benefits.

This discussion illustrates the importance of understanding the nature of the dependencies involved in a task before we can truly learn which mechanisms help teams coordinate and perform. This is particularly important for complex tasks like management decisions and large-scale software development in which dependencies are numerous and complex. For instance, our study with software teams found that there are three main types of coordination problems in this domain stemming from the specific dependencies involved: technical, temporal and software process.

Technical coordination problems arise because of inadequate management of technical dependencies among software parts that need to work well together when integrated. Temporal coordination problems arise because temporal dependencies are inadequately managed when software parts or related sub-tasks are not completed according to schedule in a synchronized manner. Software process coordination problems arise when process dependencies are inadequately managed because some team members don't adhere to the established software process or to procedures agreed upon at meetings.

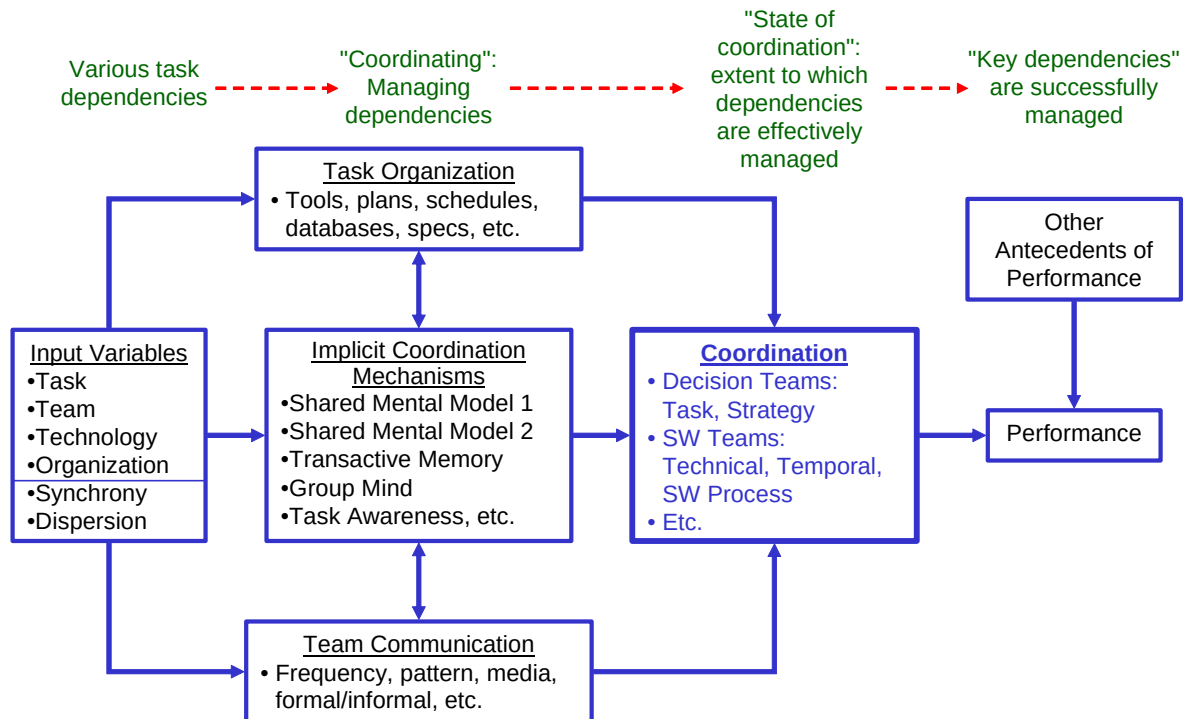


Figure 2: Reformulated Theoretical Framework

Because of the importance of understanding dependencies when studying team cognition and coordination, we have re-formulated our framework, as illustrated in Figure 2, to bring attention to the fact that different coordination mechanisms will have different effects on different

types of coordination, which in turn will have different effects on team performance—i.e., **“one size does not fit all”**. The following discussion explores the relationship between dependencies and the different elements of the framework. Once again, we start the discussion with the dependent variable of interest (i.e., team performance) and proceed backwards through the framework.

Effect of (which type of) Coordination on Performance: Key Dependencies Managed Effectively

If the purpose of a coordination mechanism is to help teams reach higher levels of performance, it is imperative that we understand not only which dependencies are present in the task, but also which ones are more critical to performance. Team cognition can only be effective in helping teams coordinate if it helps the team manage important task dependencies. Teams may be highly coordinated and still perform poorly because other antecedents of performance are weak. For example, a software team with well-coordinated task activities, but whose members have little knowledge of the application domain for which they are writing software will probably produce poor quality software (Curtis, Krasner et al. 1988). Also, some dependencies may be more important to performance than others. While many dependencies may be adequately managed, some other dependencies that are key to performance may not be managed well. For instance, a software team with high technical and temporal coordination will deliver error free software on time, but will not be considered an effective team if important features needed by clients have not been implemented.

Our research with decision teams has also shown that teams with poor task activity coordination (e.g., duplication of efforts, the left hand not knowing what the right hand is doing, unclear work assignments) don’t perform well, but it has also shown that good task activity coordination is insufficient for performance. However, we did find that good functional strategy coordination was a strong predictor of team performance because the decision task involved in this

study required the successful integration of functional strategies into a cohesive company strategy. Once again, team cognition in this case can be more effective if it involves shared knowledge of aspects of the task that are very important to performance. Also, the performance variance explained by strategy coordination was small, suggesting that other types of coordination and other variables (e.g., sound functional strategies, understanding the competition) were important too, and that the difference in performance may be affected by different types of coordination. For example, in our studies with decision teams, we found that strategy coordination affected objective financial performance but it did not directly affect performance measured with external board evaluations. The effect of strategy coordination on board evaluations was mediated by financial performance.

Effect of Coordination Mechanisms on Coordination: Managing Dependencies Effectively

Different mechanisms will have varying degrees of effectiveness in helping members manage the various types of dependencies present. Therefore, well-coordinated teams will not necessarily be those that have strong team cognition, but rather those who adopt an effective mix of mechanisms for the coordination needs of the task in which they are engaged. Consequently, when we study the effect of team cognition on coordination, it is important to answer these questions: (1) Which types of team cognition are we studying? (2) Which types of coordination are involved? And, (3) which other types of explicit coordination mechanisms are being used by the team?

Consistent with the classical view of explicit coordination, our research with decision teams shows that teams with high task coordination made heavy use of a central file sharing facility to routinely exchange documents that contain information on simulation inputs and presentations for their board of directors. They also relied on face-to-face and electronic communications for less routine aspects of the task like discussion of work assignments and key decisions to be made.

Interestingly, the same explicit coordination factors were used by teams with high strategy coordination, except for the file sharing system, probably because it did not help them integrate functional strategies, which was done mostly by communicating. Shared mental models of the task and shared mental models of the team were concurrently correlated with both task and strategy coordination, but when the proper regression models were formulated, the effect of shared mental model of the team disappeared. This suggests that once the use of all explicit coordination mechanisms is accounted for, only the shared mental model of the task has an effect on coordination in this domain (i.e. decision task, co-located and asynchronous), thus highlighting the importance of considering the effects of all coordination mechanisms used. In contrast, both shared mental models had significant and positive effects in another study with flight simulation teams (Mathieu, Goodwin et al. 2000), suggesting that managing both, task and member dependencies are important for coordination for co-located, synchronous tasks.

Similarly, experiences from our research with software teams suggest that technical personnel (e.g., software developers, testing engineers, etc.) are more concerned with technical coordination because they want to ensure that the different parts of the software they are developing integrate well. Therefore, they rely heavily on software tools (e.g., modeling tools, change management system, etc.) and formal documents (e.g., interface specifications, etc.), and communicate heavily when they encounter unusual situations or errors that need to be repaired. Technical coordination problems tend to appear when team members don't have organized shared knowledge (i.e., shared mental models) about the products being developed and the functions being implemented. In contrast, technical and project managers are more concerned with temporal and software process coordination because they want to ensure that things are done in a timely manner, per project schedules, and that things are carried out as specified by the formal software process established.

These managers rely more on planning, project management and software process documents. They also have more frequent meetings because the nature of the dependencies they have to manage is mostly non-routine. Also, process coordination problems tend to appear when managers don't have organized shared knowledge about the established software process, which leads to confusion, duplication of work, and priority conflicts, among other things.

Effect of Input Variables on Use of Coordination Mechanisms: Nature of Dependencies

When conducting research on team cognition and coordination, we need to understand how task, team and context variables influence the types of dependencies present in a task. This is important for a number of reasons. First, some findings may be limited to the particular type of task, team and/or context of the study because the nature of dependencies will be different outside of those boundaries. Second, different types of coordination mechanisms may be more effective at managing different types of dependencies. For example, team members experienced with the task and who have a long history of working together are more likely to rely on team cognition (e.g., shared mental models, transactive memory) than team members without such experience, who can only rely on task organization mechanisms and communication. Finally, some task, team and context variables cannot be changed easily (e.g., type of task, organizational culture), while some others may be more easily changed through interventions (e.g., training, new member recruitment, technology, geographic location). Consequently, it may be easier to solve some coordination problems by changing the nature of dependencies via task, team and context interventions, than by forcing teams to use different coordination mechanisms.

We will now discuss how task, team and four context variables can affect the nature of dependencies present. Two of the context variables we discuss, technology and organization, are important input variables used in team research because they have substantial effects on how teams

interact. The other two context variables, synchrony and dispersion, are essential in the study of teams mediated by information technologies because time separation (i.e., real time vs. different time) or distance separation (co-located vs. dispersed) have a substantial effect on how teams organize their work and interact (Bullen and Bennett 1993).

(a) The Task. The nature of the task really matters when studying team interaction and performance (McGrath 1991). It affects which types of dependencies are present. For example, a team assembling parts in a mechanical task will face different dependencies than a medical team with a variety of specialists trying to diagnose a patient. For instance, research on problem solving teams has shown that when team members believe that a demonstrable solution exists (i.e., an intellective task) it changes how information is sampled from team members, compared to when they believe that a demonstrable solution does not exist (i.e., a judgment task) (Stasser and Stewart 1992).

(b) The Team. Many team variables can affect the nature of dependencies present. For example, how long or how often the team has worked or trained together in the past, the team composition (e.g., homogeneity, expertise, etc.), member acquisition practices (e.g., labor markets, socialization, etc.), and team size will affect team cognition development (Cannon-Bowers, Salas et al. 1993; Klimoski and Mohammed 1994; Levesque, Wilson et al. 2001; Rentsch and Klimoski 2001). Teams that have worked together for a long time may have well-developed team cognition mechanisms and work practices that help them minimize dependencies. Team variables cannot only affect team cognition development, but also influence which explicit coordination mechanisms its members employ. For example, the number and complexity of dependencies in software development increases exponentially as the size of the team increases and it makes it difficult to coordinate and communicate (Brooks

1995). So, larger teams are more likely to rely on task organization mechanisms (e.g., software tools, manuals, specifications, schedules, etc.), while smaller teams are more likely to rely on team communication (VanDeVen, Delbecq et al. 1976). Other studies with software teams have also found that project size and task uncertainty make coordination more difficult (Kraut and Streeter 1995; Espinosa 2002), but that teams that employ formal routines reduce some of the resulting negative effects (Sproull and Kiesler 1991).

(c) Context: Technology. Communications and other technologies can affect how teams interact (McGrath and Hollingshead 1994). For example, a study of relationships between technologists and radiologists at two hospitals found that new technologies changed the nature of individual roles, which in turn had a substantial effect on how technologists and radiologists interacted (Barley 1990). Other research studies have also suggested that information technologies affect dependencies, information flow, and workflow among collaborators (Sproull and Kiesler 1991; Grinter, Herbsleb et al. 1999). For instance, smaller software projects don't always utilize configuration management systems, which creates member dependencies when different developers need to work on the same part of the software code. In contrast, large-scale software projects use such systems, which enable multiple developers to work on the same part of the software code simultaneously without having to interact with each other (i.e., the system handles this for them), which reduces member dependencies.

(d) Context: Organization. Organizational factors (e.g., culture, structure, standard procedures) also affect how teams interact. In fact, some research studies suggest that organizations are social systems that affect (and are affected by) how technologies are used (Orlikowski 1992; DeSanctis and Poole 1994; Orlikowski 1996), all of which can affect the types of dependencies present in a task.

(e) Context: Synchrony and Geographic Dispersion. Teams can operate synchronously (i.e., same time or real time) or asynchronously (i.e., different time) and can be either co-located (i.e., same place) or geographically dispersed (i.e., different place), thus creating four possible modes of team interaction that need to be considered when researching teams (Bullen and Bennett 1993). Consequently, synchrony and geographic dispersion need to be considered when studying team cognition because they can generate different work arrangements with different resulting sets and types of dependencies. Asynchronous and geographically dispersed teams have fewer opportunities to interact, communicate less spontaneously, and use leaner communication media (e.g., electronic mail, telephone). Consequently, time and distance separation will not only make it more difficult for team members to develop and use shared cognition to coordinate, but it will also affect the mix of explicit and implicit mechanisms that they will employ to coordinate. So, while research in same-time/same-place contexts has yielded very useful theoretical foundations and empirical support for the positive effects of team cognition (Liang, Moreland et al. 1995; Stout, Cannon-Bowers et al. 1999; Mathieu, Goodwin et al. 2000), we don't know if these findings extend to different-time and different-place contexts.

Fortunately, as we discussed earlier, new empirical research studies are beginning to emerge in different time-same place contexts (i.e., asynchronous, co-located), which are beginning to show evidence of the positive effects of team cognition on performance (Crowston and Kammerer 1998; Faraj and Sproull 2000; Lewis 2000; Espinosa 2002; Espinosa, Carley et al. 2002) and helping to identify antecedents of team cognition (Levesque, Wilson et al. 2001; Rentsch and Klimoski 2001). Our study with decision teams found a positive effect of the shared mental model of the task, but found no effect of the shared mental model of the team.

Our explanation for this is that the shared mental model of the team is perhaps important in high-paced, real-time tasks like a flight simulation because this shared model helps team members explain and anticipate each other's actions, which helps manage member dependencies, which are key to performance in these contexts. Conversely, management decision teams are asynchronous, so having shared knowledge of the task is perhaps more important for managing the respective task dependencies than having a well-developed shared mental model of the team.

However, there is almost no empirical research in different place contexts (i.e., geographically dispersed). Our research studies try to fill this gap by focusing specifically on different place and/or different time contexts. For example, our research with software teams (i.e., asynchronous, dispersed) found that dependencies and coordination needs are very different for co-located than for dispersed teams, but that having organized shared knowledge about key concepts, processes and products (i.e., shared mental models), and knowing who knows what in the team (i.e., transactive memory) helped teams coordinate, thus reducing some of the coordination problems associated with geographic distance. We also found that software development time was reduced when team members have shared familiarity with prior software projects and with the same parts of the software code, and that this effect was stronger for geographically distributed teams than for co-located teams.

As discussed previously, research in team cognition can benefit by taking into account the task context, particularly synchrony and geographic dispersion. Research findings are likely to differ depending on time and distance separation because the nature of the dependencies present will change. In fact, interventions that change the synchrony or dispersion of a task can have substantial effects on coordination. For instance, while a field study of global software teams

found that there are more complex dependencies and coordination challenges when development is done across sites than when it is done at a single site (Herbsleb, Mockus et al. 2001), a related prior study also found that large-scale software development is often distributed in ways that reduce dependencies across sites, such as: (a) develop all related software functions (e.g., call handling, billing) at a single site; (b) develop software for similar parts of the software product (e.g., base station controllers) at a single site; (c) keep similar software development phases (e.g., testing) at a single site; (d) develop a core product at a single site and customize it for specific client needs at a site near the client (Grinter, Herbsleb et al. 1999).

Conclusions

Our arguments and examples in this chapter bring attention to the fact that it is important to consider explicit coordination mechanisms and task dependencies when studying team cognition. Results from empirical studies in team cognition will be more useful if they limit the scope and validity of the study to particular contexts involving the use of a given set of explicit coordination mechanisms. At the same time, if the use of explicit mechanisms vary across teams, it is important to control for the use of such mechanisms. This underscores the importance of using an integrated research framework that incorporates both implicit and explicit coordination mechanisms. It is also important to understand how task, team and context variables affect work arrangements and task dependencies when we study team coordination. Empirical findings from research studies about the effect of a particular team cognition construct on team coordination and performance will be more useful for research and practice if the specifics of the task dependencies are articulated and considered. Also, while coordination may not be a sufficient condition for effective team performance, it is certainly necessary for tasks with substantial dependencies. At the same time,

teams that are highly coordinated in certain aspects of the task may not necessarily exhibit high performance levels unless other antecedents of performance are also present.

There is an increased interest these days in how different types of team cognition can influence team coordination and performance. Many team cognition constructs have been identified in the literature (Mohammed and Dumville 2001) and, while these constructs are conceptually distinct, they share some similarities. In particular, most constructs involve some form of compatible, complementary or similar knowledge among team members (Cannon-Bowers and Salas 2001). This team knowledge, in turn, arguably helps team members coordinate their actions with the rest of the team and communicate more effectively. Although such concepts and theories are intuitive and useful in helping us explain team coordination and performance, we still need more empirical evidence. Through our research, we have identified a set of specific areas and issues that need further development to better understand team cognition:

1. There is very little agreement or consistency in the literature about how to measure shared cognition (Cooke, Salas et al. 2000; Lewis 2000; Mohammed and Dumville 2001). Further research and more consistency of methods are needed in this area to be able to compare results across studies.
2. There is very little empirical work supporting these theories and concepts. More empirical work is necessary in this area (Mathieu, Goodwin et al. 2000; Mohammed and Dumville 2001). Part of the reason for this has to do with the difficulty of measuring shared cognition.
3. There is not much discussion in the literature about how the effects of team cognition may vary depending on the type of task involved. The nature of the task and its context matter when studying team performance (McGrath 1991). More specifically, time and distance separation make a difference when studying team coordination (Bullen and Bennett 1993).

4. How team cognition affects coordination and performance will depend on the extent to which teams use certain explicit coordination mechanisms like collaboration tools, electronic communication, shared databases, and division of labor. More research is needed in which the effects of explicit and implicit coordination mechanisms are jointly explored.
5. Complex tasks have multiple dependencies. The management specific task dependencies will require different types of coordination. Therefore, more research is needed to better understand: (a) the various effects that specific explicit and implicit coordination mechanisms have on the effective management of different task dependencies; and (b) how different types of coordination affect different aspects of team performance.

The first point about measures has been partly addressed in other studies (Cooke, Salas et al. 2000; Espinosa 2001; Espinosa and Carley 2001). In this chapter we have addressed the remaining four issues by focusing on two overarching themes of concern for team cognition research. The first one is the need for a research framework that incorporates explicit mechanisms (from the classical organizational research literature) and implicit coordination mechanisms (from more recent research in team cognition). The second overarching theme has to do with the need to better understand the nature of the multiple dependencies and coordination types involved in a task (i.e., one size does not fit all). Complex tasks often have many types of dependencies, some more critical than others, and certain coordination mechanisms may be more effective than others for particular types of dependencies. Conversely, one particular mechanism (e.g. shared mental model of the task) may not be equally effective in managing different dependencies. Consequently, an effective strategy for coordination success involves finding a mix of coordination mechanisms well suited for the task. Furthermore, this mix will probably need to change as the task progresses over time (e.g., communication may be more effective at the beginning of the task while division of

labor or shared mental models may be more effective later on). This mix will also need to change with distance separation (i.e., co-located vs. distributed).

References

- Barley, S. (1990). "The Alignment of Technology and Structure Through Roles and Networks." Administrative Science Quarterly 35(March): p.61-103.
- Brooks, F. (1995). The Mythical Man-Month: Essays on Software Engineering, A. Wesley.
- Bullen, C. and J. Bennett (1993). Groupware in Practice: An Interpretation of Work Experiences, in Groupware and Computer-Supported Cooperative Work: Assisting Human-Human Collaboration, R. Baecker (Eds.). CA, Morgan Kaufman Publishers: p.69-84.
- Cannon-Bowers, J. A. and E. Salas (2001). "Reflections on Shared Cognition." Journal of Organizational Behavior 22(2): p.195-202.
- Cannon-Bowers, J. A., E. Salas and S. Converse (1993). Shared Mental Models in Expert Team Decision-Making, in Individual and Group Decision-Making: Current Issues, J. Castellan (Eds.), LEA Publishers: p.221.
- Cooke, N. J., E. Salas, J. A. Cannon-Bowers and R. J. Stout (2000). "Measuring Team Knowledge." Human Factors 42(1): p.151-173.
- Cramton, C. D. (2001). "The Mutual Knowledge Problem and Its Consequences for Dispersed Collaboration." Organization Science 12(3): p.346-371.
- Crowston, K. and E. E. Kammerer (1998). "Coordination and Collective Mind in Software Requirements Development." IBM Systems Journal 37(2): p.227-245.
- Curtis, B., H. Krasner and N. Iscoe (1988). "A Field Study of the Software Design Process for Large Systems." Communications of the ACM 31(11): p.1268-1286.
- DeSanctis, G. and M. S. Poole (1994). "Capturing the Complexity in Advanced Technology Use: Adaptive Structuration Theory." Organization Science 5(2): p.121-145.
- Endsley, M. (1995). "Toward a Theory of Situation Awareness in Dynamic Systems." Human Factors 37(1).
- Entin, E. E. and D. Serfaty (1999). "Adaptive Team Coordination." Journal of Human Factors 41(2): p.321-325.
- Espinosa, J. A. (2001). Shared Mental Models: Accuracy and Visual Representation. Americas Conference in Information Systems, Boston MA, Association for Information Systems
- Espinosa, J. A. (2002). Shared Mental Models and Coordination in Large-Scale, Distributed Software Development, doctoral dissertation, Carnegie Mellon University, Pittsburgh. UMI 3065743, ISBN 0-493-85125-9
- Espinosa, J. A. and K. Carley (2001). Measuring Team Mental Models. Academy of Management Conference, Washington D.C.
- Espinosa, J. A., K. Carley, R. E. Kraut, F. J. Lerch and S. Fussell (2002). The Effect of Task Knowledge Similarity and Distribution on Asynchronous Team Coordination and Performance: Empirical Evidence from Decision Teams. Information Systems Cognitive Research Exchange (IS CoRE) workshop, Barcelona, Spain
- Espinosa, J. A., R. E. Kraut, F. J. Lerch, S. A. Slaughter, J. D. Herbsleb and A. Mockus (2001). Shared Mental Models and Coordination in Large-Scale, Distributed Software Development. International Conference in Information Systems, New Orleans, LA
- Espinosa, J. A., R. E. Kraut, S. A. Slaughter, F. J. Lerch, J. D. Herbsleb and A. Mockus (2002). Shared Mental Models, Familiarity, and Coordination: A

- Multi-Method Study of Distributed Software Teams. International Conference in Information Systems, Barcelona, Spain
- Faraj, S. and L. Sproull (2000). "Coordinating Expertise in Software Development Teams." Management Science 46(12): p.1554-1568.
- Fussell, S. and R. Krauss (1992). "Coordination of Knowledge in Communication: Effects of Speakers' Assumptions About What Other Know." Journal of Personality and Social Psychology 62(3): p.378-391.
- Grinter, R. E., J. D. Herbsleb and D. E. Perry (1999). The Geography of Coordination: Dealing with Distance in R&D Work. International ACM SIGGROUP Conference on Supporting Group Work (Group 99), Phoenix, Arizona, ACM Press
- Herbsleb, J., A. Mockus, T. Finholt and R. E. Grinter (2001). An Empirical Study of Global Software Development: Distance and Speed. 23rd. International Conference on Software Engineering (ICSE), Toronto, Canada, IEEE Computer Society Press
- Kleinman, D. L. and D. Serfaty (1989). Team Performance Assessment in Distributed Decision-Making. Interactive Networked Simulation for Training Conference, Orlando, FL., Naval Training System Center
- Klimoski, R. J. and S. Mohammed (1994). "Team Mental Model: Construct or Methaphor." Journal of Management 20(2): p.403-437.
- Kraemer, K. and A. Pinsonneault (1990). Technology and Groups: Assessment of the Empirical Research, in Intellectual Teamwork: Social and Technological Foundations of Cooperative Work, J. Galegher, R. E. Kraut and C. Egidio (Eds.), LEA Publishers.
- Kraiger, K. and L. Wenzel (1997). Conceptual Development and Empirical Evaluation of Measures of Shared Mental Models as Indicators of Team Effectiveness, in Team Performance Assessment and Measurement, M. Brannick, E. Salas and C. Prince (Eds.), LEA Publishers.
- Krauss, R. and S. Fussell (1990). Mutual Knowledge and Communicative Effectiveness, in Intellectual Teamwork: Social and Technological Foundations of Cooperative Work, J. Galegher, R. E. Kraut and C. Egidio (Eds.). Hillsdale, NJ, Lawrence Erlbaum: p.111-146.
- Kraut, R. E. and L. A. Streeter (1995). "Coordination in Software Development." Communications of the ACM 38(3): p.69-81.
- Levesque, L. L., J. M. Wilson and D. R. Wholey (2001). "Cognitive Divergence and Shared Mental Models in Software Development Project Teams." Journal of Organizational Behavior 22(2): p.135-144.
- Lewis, K. (2000). Transactive Memory and Performance of Management Consulting Teams: Examining Construct and Predictive Validity of a New Scale. Academy of Management Conference, Toronto
- Liang, D., R. Moreland and L. Argote (1995). "Group Versus Individual Training and Group Performance: The Mediating Role of Transactive Memory." Personality and Social Psychology Bulletin 21.
- Malone, T. and K. Crowston (1990). What is Coordination Theory and How Can it Help Design Cooperative Work Systems. Computer Supported Collaborative Work, ACM Press
- Malone, T. and K. Crowston (1994). "The Interdisciplinary Study of Coordination." ACM Computing Surveys 26(1): p.87-119.
- March, J. and H. Simon (1958). Organizations, John Wiley and Sons.
- Mathieu, J., G. F. Goodwin, T. S. Heffner, E. Salas and J. A. Cannon-Bowers (2000). "The Influence of Shared Mental Models on Team Process and Performance." Journal of Applied Psychology 85(2): p.273-283.
- McGrath, J. (1991). "Time, Interaction and Performance (TIP)." Small Group Research 22.
- McGrath, J. and A. Hollingshead (1994). Groups Interacting With Technology, Sage Publications.

- Mohammed, S. and B. C. Dumville (2001). "Team Mental Models in a Team Knowledge Framework: Expanding Theory and Measurement Across Disciplinary Boundaries." Journal of Organizational Behavior 22(2): p.89-106.
- Orlikowski, W. (1992). Learning from Notes: Organizational Issues in Groupware Implementation. Computer Supported Collaborative Work, ACM Press
- Orlikowski, W. (1996). "Improvising Organizational Transformation Over Time: A Situated Change Perspective." Information Systems Research(March).
- Rentsch, J. R. and R. J. Klimoski (2001). "Why do Great Minds Think Alike?: Antecedents of Team Member Schema Agreement." Journal of Organizational Behavior 22(2): p.107-120.
- Rouse, W. B., J. A. Cannon-Bowers and E. Salas (1992). "The Role of Mental Models in Team Performance in Complex Systems." IEEE Transactions on Systems, Man & Cybernetics 22: p.1296-1308.
- Rouse, W. B. and N. M. Morris (1986). "On Looking Into the Black Box: Prospects and Limits in the Search for Mental Models." Psychological Bulletin 100: p.349-363.
- Sproull, L. and S. Kiesler (1991). Connections: New Ways of Working in the Networked Organization, MIT Press.
- Stasser, G. and D. Stewart (1992). "Discovery of Hidden Profiles by Decision-Making Groups: Solving a Problem Versus Making a Judgement." Journal of Personality and Social Psychology 63(3): p.426-434.
- Stout, J. A., J. Cannon-Bowers and E. Salas (1999). "Planning, Shared Mental Models, and Coordinated Performance: An Empirical Link is Established." Human Factors 41: p.61-71.
- Thompson, J. (1967). Organizations in Action, McGraw-Hill.
- VanDeVen, A. H., L. A. Delbecq and R. J. Koenig (1976). "Determinants of Coordination Modes Within Organizations." American Sociological Review 41(April): p.322-338.
- Wegner, D. (1986). Transactive Memory: A Contemporary Analysis of the Group Mind, in Theories of Group Behavior, B. Mullen and G. Goethals (Eds.), Springer-Verlag.
- Weick, K. and K. Roberts (1993). "Collective Mind in Organizations: Heedful Interrelating on Flight Decks." Administrative Science Quarterly 38(3): p.357-381.
- Wellens, R. (1993). Group Situation Awareness and Distributed Decision-Making: From Military to Civilian Applications, in Individual and Group Decision-Making: Current Issues, J. Castellan (Eds.), LEA Publishers: p.267.
- Wholey, D., S. Kiesler and K. Carley (1996). "Learning Teamwork: Emergence of Organization in Novice Software Development Teams." Working paper.
- Wittenbaum, G. M. and G. Stasser (1996). Management of Information in Small Groups, in What's Social about Social Cognition?, J. L. Nye and A. M. Brower (Eds.), Sage Publications: p.3.